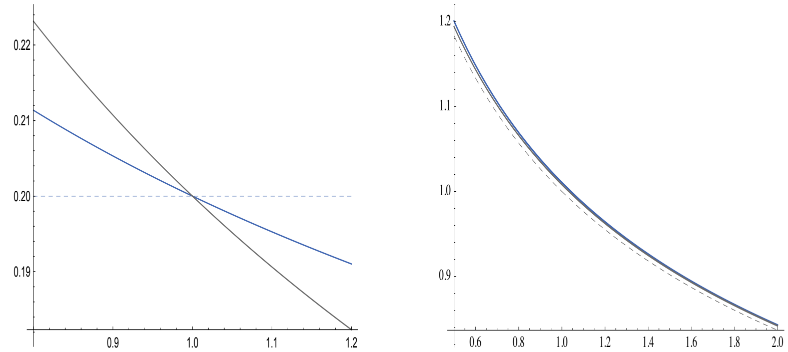

Figure 1: On the left we have plotted the leading-order asymptotic implied volatility $\hat{\sigma}_0(K)$ of a European put or call option as the maturity $T \to 0$, as a function of the strike K, for the Black–Scholes case $\beta = 1$ (dashed), $\beta = \frac{1}{2}$ (blue), and $\beta = 0$ (grey), with $\sigma = .2$ and $S_0 = 1$. On the right we have plotted $\hat{\sigma}_0(K)$ (dashed), the first-order approximation $\hat{\sigma}_0(K) + \hat{\sigma}_1(K)T$ from (4) (blue), and the answer obtained by numerically integrating the density of S_T (grey), with $\beta = \frac{1}{2}$, $\sigma = 1$, $r = 0$, and $T = 1$. Even for the relatively large value of σ used here, the smile changes little as T increases, compared with typical Markovian or rough stochastic-volatility models and jump models.

1 Introduction to Project 2: Deep Pricing in the CEV Model

1.1 Model description and no-arbitrage

As in the project description, we assume S follows the CEV process

$$dS_t \quad = \quad \mu S_t \, dt + \sigma S_t^{\beta} \, dW_t$$

under the real-world probability measure $\mathbb{P}$, where $0 \le \beta \le 1$ and W is a standard Brownian motion. We work with the non-negative solution which is absorbed when it first hits zero. Under the risk-neutral measure $\mathbb{Q}$, the no-arbitrage drift is rS_t, so

$$dS_t \quad = \quad rS_t \, dt + \sigma S_t^{\beta} \, dW_t.$$

The discounted stock $\widetilde{S}_t = e^{-rt} S_t$ must be a martingale under $\mathbb{Q}$ to exclude arbitrage. In the project document the CEV exponent is denoted by γ, but most authors use β. The model reduces to the usual Black–Scholes model when $\beta = 1$.

1.2 Properties of the CEV model

The process S is a **one-dimensional diffusion process** and hence a **Markov process**. In particular, for $0 \le u \le t$ and every bounded measurable function f,

$$\mathbb{E}(f(S_t) \mid \mathcal{F}_u^W) \quad = \quad \mathbb{E}(f(S_t) \mid S_u).$$

The CEV model gives rise to a decreasing **implied-volatility skew**.[1][2] The skew becomes more pronounced as β decreases; see Figure 1. This makes the model particularly relevant for equity and equity-index options, although it does not reproduce the pronounced U-shaped smiles often observed in foreign-exchange markets.

Unlike the Black–Scholes model, S can **hit zero** with non-zero probability when $\beta < 1$.[3] We impose **absorption at zero**. This is also financially natural: a non-negative discounted stock which is known to leave zero and become strictly positive again cannot be a martingale.

Let $p = 1 - \beta > 0$, and define the regularized upper incomplete gamma function by

$$G(a, x) \quad = \quad \frac{1}{\Gamma(a)} \int_x^{\infty} e^{-u} u^{a-1} \, du.$$

For $r > 0$, the probability of absorption by time T is

$$\mathbb{Q}(S_T = 0) \quad = \quad G\Big(\frac{1}{2p}, \frac{rS_0^{2p}}{\sigma^2 p(1 - e^{-2rpT})}\Big) \tag{1}$$

[1] For an option price lying strictly inside the corresponding no-arbitrage bounds and with positive maturity, implied volatility is the unique positive solution σ_{imp} of $P^{\text{BS}}(S, K, \sigma_{\text{imp}}, T, r) = P$.

[2] When $\beta < 1$, the proportional volatility is $\sigma S_t^{\beta - 1}$, which increases as S_t decreases. See "Pricing Options on Scalar Diffusions: An Eigenfunction Expansion Approach" by Linetsky for more on the CEV model.

[3] When $\beta = 1$, $S_t > 0$ for every finite t, although $S_t \to 0$ as $t \to \infty$ if $r - \frac{1}{2}\sigma^2 < 0$.

For $r = 0$, this becomes

$$\mathbb{Q}(S_T = 0) \quad = \quad G(\frac{1}{2p}, \frac{S_0^{2p}}{2\sigma^2 p^2 T}).$$

The absorption probability increases as σ increases and decreases as r increases. For completeness, if one instead takes $\beta > 1$ and $r = 0$, then S is a **strict local martingale** and hence a strict supermartingale; this case is excluded from the project.

1.3 Terminal stock-price density and pricing European options

For $r = 0$ and $\tau > 0$, the density of the positive part of S_τ under $\mathbb{Q}$ is

$$p(\tau, S_0, S) \quad = \quad \frac{S^{-2\bar{\beta}-\frac{3}{2}} S_0^{\frac{1}{2}}}{\sigma^2 |\bar{\beta}| \tau} \exp(-\frac{S_0^{-2\bar{\beta}} + S^{-2\bar{\beta}}}{2\sigma^2 \bar{\beta}^2 \tau}) I_\nu(\frac{S_0^{-\bar{\beta}} S^{-\bar{\beta}}}{\sigma^2 \bar{\beta}^2 \tau}), \qquad S > 0, \tag{2}$$

where $\bar{\beta} = \beta - 1$, $\nu = 1/(2|\bar{\beta}|)$, and I_ν is the **modified Bessel function of the first kind**. MATLAB, Mathematica, and Python all have built-in implementations of this function. The density in (2) does not integrate to one because of the atom at zero, so it is called a *defective density*.

For $r > 0$, define

$$q(\tau) \quad = \quad \frac{1 - e^{-2r(1-\beta)\tau}}{2r(1-\beta)} \quad = \quad \frac{e^{2r\bar{\beta}\tau} - 1}{2r\bar{\beta}}.$$

The density of the positive part of S_τ is then

$$f_\tau(S) \quad = \quad e^{-r\tau} p(q(\tau), S_0, S e^{-r\tau}), \qquad S > 0.$$

Consequently, the time-zero price of a European call is

$$C(S_0, K, \tau) \quad = \quad e^{-r\tau} \mathbb{E}^{\mathbb{Q}} (S_\tau - K)^+ \quad = \quad e^{-r\tau} \int_K^\infty (S - K) f_\tau(S)\, dS. \tag{3}$$

A European put can be priced directly by including the atom at zero:

$$P(S_0, K, \tau) \quad = \quad e^{-r\tau} \{K \mathbb{Q}(S_\tau = 0) + \int_0^K (K - S) f_\tau(S)\, dS\}.$$

Equivalently, one can use put–call parity,

$$C(S_0, K, \tau) \, + \, K e^{-r\tau} \quad = \quad P(S_0, K, \tau) \, + \, S_0.$$

A closed-form CEV call formula was derived by Cox in 1975. Once the European option price has been computed, its implied volatility can be obtained by numerically inverting the Black–Scholes formula.

For a general one-dimensional local-volatility model

$$dS_t \quad = \quad r S_t\, dt + S_t a(S_t)\, dW_t,$$

with a sufficiently regular, the small-time implied-volatility expansion is

$$\hat{\sigma}(K, T) \quad = \quad \hat{\sigma}_0(K) + \hat{\sigma}_1(K) T + O(T^2),$$

where

$$\hat{\sigma}_0(K) \quad = \quad \frac{\log(K/S_0)}{\int_{S_0}^K \frac{du}{u a(u)}}$$

and

$$\hat{\sigma}_1(K) \quad = \quad \frac{\hat{\sigma}_0(K)^3}{\log^2(K/S_0)} (\log \frac{\sqrt{a(S_0) a(K)}}{\hat{\sigma}_0(K)} + r \int_{S_0}^K \{\frac{1}{a(u)^2} - \frac{1}{\hat{\sigma}_0(u)^2}\} \frac{du}{u}). \tag{4}$$

where $a(u) = \sigma u^{\beta - 1}$ for the CEV model. The second plot in Figure 1 gives a numerical example.

For $r > 0$, the positive-part transition density also admits a **spectral expansion**

$$p(\tau, S_0, S) \quad = \quad m(S) \sum_{n=1}^\infty e^{-\lambda_n \tau} \varphi_n(S_0) \varphi_n(S),$$

which is conceptually similar to a Fourier series. Here

$$m(S) \quad = \quad \frac{2}{\sigma^2 S^{2\beta}} \exp(\frac{r S^{-2\bar{\beta}}}{\sigma^2 |\bar{\beta}|})$$

is the speed density, and the eigenfunctions are orthonormal in the sense that

$$\int_0^\infty \varphi_n(S) \varphi_m(S) m(S)\, dS \quad = \quad \begin{cases} 1, & n = m, \\ 0, & n \neq m. \end{cases}$$

1.4 American put options

Consider an American put at time t, where $0 \le t \le T$. It can be exercised at any stopping time τ satisfying $t \le \tau \le T$, in which case its owner receives $(K - S_\tau)^+$ at time τ. Its no-arbitrage price is

$$P(S,t) \quad = \quad \sup_{\tau \in \mathcal{T}_{t,T}} \mathbb{E}^{\mathbb{Q}}(e^{-r(\tau - t)}(K - S_\tau)^+ \mid S_t = S),$$

where $\mathcal{T}_{t,T}$ is the collection of stopping times taking values between t and T. Recall that τ is a stopping time if $\{\tau \le u\} \in \mathcal{F}_u$ for every u. A classic example of a random time which is not a stopping time is the last time at which W hits zero before T. See, for example, the books by Peskir and Shiryaev and by Huyên Pham for more theoretical background on optimal stopping.

Remark 1.1 Since the model is time-homogeneous, $P(S,t)$ is equal to the time-zero price of an American put with maturity $T - t$. This will be important for Part 2.

Since the deterministic stopping time T is admissible,

$$P(S,t) \quad \ge \quad e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}((K - S_T)^+ \mid S_t = S).$$

The right-hand side is the CEV European put price, which can be computed from the CEV transition density or from the closed-form CEV call formula and put–call parity.

The price of a European option under the CEV model satisfies the CEV pricing PDE

$$P_t + rSP_S + \frac{1}{2}\sigma^2 S^{2\beta} P_{SS} \quad = \quad rP$$

with terminal condition $P(S,T) = (K - S)^+$. The American put price satisfies the variational inequality

$$\max\{P_t + rSP_S + \frac{1}{2}\sigma^2 S^{2\beta} P_{SS} - rP, (K - S)^+ - P\} \quad = \quad 0. \tag{5}$$

Equivalently, there is an **optimal exercise boundary** $B(t)$ such that

$$\begin{aligned} P_t + rSP_S + \frac{1}{2}\sigma^2 S^{2\beta} P_{SS} &= rP, \qquad S > B(t), \\ P(S,t) &= K - S, \qquad 0 \le S \le B(t), \\ P(S,T) &= (K - S)^+, \\ P(S,t) &\ge (K - S)^+, \qquad S \ge 0. \end{aligned} \tag{6}$$

This is a **free-boundary problem**, also called an *obstacle problem*, because the boundary B is unknown and forms part of the solution. A European put can lie below its intrinsic value, but the American put cannot. If $S_t \le B(t)$, it is optimal to exercise immediately.

The value-matching and **smooth-fit** conditions are

$$\begin{aligned} P(B(t),t) &= K - B(t), \\ P_S(B(t),t) &= -1. \end{aligned}$$

Value matching gives continuity of the price across the exercise boundary, while smooth fit gives continuity of its first derivative with respect to S.

Remark 1.2 The free-boundary problem cannot be solved explicitly in general, but it can be solved numerically using a finite-difference scheme, for example an explicit or implicit scheme; see FM06 for further details.

For the **explicit central-difference scheme**, let $S_i = i\Delta S$, $t_n = n\Delta t$, and write P_i^n for the approximation to $P(S_i, t_n)$. Starting from the terminal values $P_i^N = (K - S_i)^+$, march backwards using

$$P_i^{n-1} \quad = \quad \max\{a_i P_{i-1}^n + b_i P_i^n + c_i P_{i+1}^n, (K - S_i)^+\}, \tag{7}$$

for $n = N, N-1, \ldots, 1$, where

$$\begin{aligned} a_i &= \Delta t\{\frac{\sigma^2 S_i^{2\beta}}{2(\Delta S)^2} - \frac{rS_i}{2\Delta S}\}, \\ b_i &= 1 - \Delta t\{\frac{\sigma^2 S_i^{2\beta}}{(\Delta S)^2} + r\}, \\ c_i &= \Delta t\{\frac{\sigma^2 S_i^{2\beta}}{2(\Delta S)^2} + \frac{rS_i}{2\Delta S}\}. \end{aligned}$$

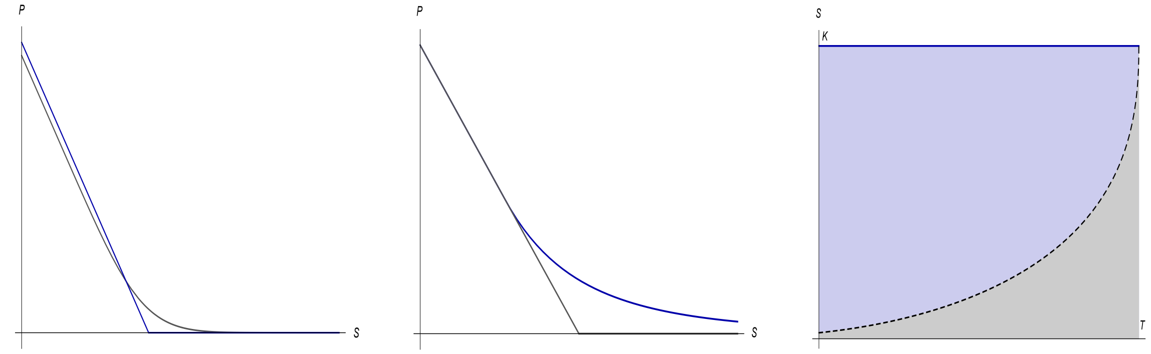

Figure 2: In the left-hand plot we have plotted the value of a European put against the initial stock price. In the middle plot we show the value of an American put as a function of the initial stock price; below a critical stock price, the American put is equal to its intrinsic value $(K-S)^+$. On the right we have plotted the small-time approximation to the early-exercise boundary $B(t)$. The blue area is the continuation region and the grey area is the exercise, or stopping, region.

Use the boundary conditions $P_0^n = K$ and $P_M^n = 0$, with $S_M = S_{\max} > K$ chosen sufficiently large. For monotonicity and stability one should impose $a_i \geq 0$, $b_i \geq 0$, and $c_i \geq 0$ for every interior grid point. In particular,

$$\Delta t\{\frac{\sigma^2 S_{\max}^{2\beta}}{(\Delta S)^2} + r\} \quad \leq \quad 1$$

is a convenient sufficient restriction for $b_i \geq 0$. The crude condition $\Delta t/(\Delta S)^2 < 1$ is not sufficient. If $a_i < 0$ near zero, replace the central approximation of the drift derivative by an upwind approximation.

The explicit scheme is easy to implement, although a fine spatial grid can require a very large number of time steps. Implicit methods for American options require the solution of a linear complementarity problem, commonly by a projected successive over-relaxation method. The explicit scheme can also be extended to Markovian, non-rough stochastic-volatility models, including models with non-zero correlation.

You then need to design and train a neural network to learn the pricing map P.

1.5 Outline of tasks to perform for Part 2

First write an explicit finite-difference scheme to compute $P(S_0, K, T, r, \sigma, \beta)$ for the test parameters below, and check that your answer agrees with the stated value. As a sanity check, test the same finite-difference implementation for European options and compare with CEV Monte Carlo estimates, or test the Black–Scholes case against the Black–Scholes formula. Ordinary forward Monte Carlo does not by itself solve the American optimal-stopping problem; a method such as least-squares Monte Carlo would be needed for a Monte Carlo comparison.

Then compute the American put value for a range of values of r and β. It is usually better to sample the parameters randomly from realistic non-degenerate ranges, for example

$$.5 \le S_0 \le 1.5, \qquad .5 \le K \le 1.5, \qquad .01 \le \sigma \le .5, \qquad .01 \le T \le 1, \qquad 0 \le r \le .10, \qquad 0 \le \beta \le 1.$$

Adjust $S_{\min}$ and $S_{\max}$ intelligently as functions of the parameters, especially σ, or choose a grid wide enough to remain accurate over the full parameter range. Test convergence by refining the time and space grids while preserving the coefficient conditions following (7).

As observed in the project document, one finite-difference run produces values at all intermediate stock-grid points and time levels. Since the model is time-homogeneous, the value $P(S, t)$ of a put with maturity T is the same as the time-zero value of a put with maturity $T - t$. Thus one run automatically produces training values for a range of S_0 and maturity values. Store these values in a large text file to avoid recomputing them later.

The scaling relation is

$$P(S_0, K, T, r, \sigma, \beta) \quad = \quad \frac{1}{\lambda} P(\lambda S_0, \lambda K, \alpha T, \frac{r}{\alpha}, \lambda^{1-\beta} \alpha^{-\frac{1}{2}} \sigma, \beta). \tag{8}$$

This allows prices for new values of K, T, r, and σ to be inferred from existing finite-difference runs. Equivalent forms include

$$\begin{aligned}
P(S_0, K/\lambda, T, r, \sigma, \beta) &= \frac{1}{\lambda} P(\lambda S_0, K, \alpha T, \frac{r}{\alpha}, \lambda^{1-\beta}\alpha^{-\frac{1}{2}}\sigma, \beta), \\
\lambda P(S_0, K/\lambda, T, r, \frac{\sigma}{\lambda^{1-\beta}}, \beta) &= P(\lambda S_0, K, \alpha T, \frac{r}{\alpha}, \alpha^{-\frac{1}{2}}\sigma, \beta), \\
\lambda P(S_0, K, T, \alpha r, \alpha^{\frac{1}{2}}\sigma, \beta) &= P(\lambda S_0, \lambda K, \alpha T, r, \lambda^{1-\beta}\sigma, \beta), \\
\lambda P(S_0, K/\lambda, T, \alpha r, \frac{\alpha^{\frac{1}{2}}\sigma}{\lambda^{1-\beta}}, \beta) &= P(\lambda S_0, K, \alpha T, r, \sigma, \beta).
\end{aligned} \tag{9}$$

The last form is particularly useful for Part 2. If λS_0 or αT is not a grid point, use linear interpolation from nearby grid values.

Test parameters: $\beta = .7$, $S_0 = 1$, $\sigma = .2$, $T = 1$, $K = .95$, and $r = .02$. With 200,000 time steps and 400 spatial steps, the American put price is approximately .0489. Now take $\lambda = 1.03$ and $\alpha = 1.02$, multiply S_0 and K by λ, multiply T by α, replace r by r/α, and replace σ by

$$\lambda^{1-\beta}\alpha^{-\frac{1}{2}}\sigma.$$

Reprice the American put and divide the resulting price by λ. The answer should again be .0489 to three significant figures. This numerically checks (8) for these parameters.

Remark 1.3 For $r > 0$, the absolute local-volatility coefficient of the CEV model at the strike is σK^β. Define

$$\eta \quad = \quad \frac{\sigma^2 K^{2\beta}}{8\pi (rK)^2} = \frac{\sigma^2 K^{2\beta-2}}{8\pi r^2}.$$

The optimal exercise boundary has the following close-to-expiry behaviour:

$$B(t) \quad = \quad K - \sigma K^\beta \{\tau \log(\frac{\eta}{\tau}\{1 + \frac{1}{\log(\tau/\eta)} + O(\frac{1}{\log^3(\tau/\eta)})\}^{-2})\}^{\frac{1}{2}}, \tag{10}$$

where $\tau = T - t$. In particular, $B(t) \to K$ as $t \to T$. See Figure 3 and the 2017 *Risk* article by De Marco and Henry-Labordère for further details.

Possible ideas for Part 3 include: evaluating a CEV process at an independent stochastic time in order to generate a U-shaped implied-volatility smile, using the spectral expansion discussed above; pricing and **calibrating** European and VIX options under **rough-volatility** models, such as rough Bergomi, rough Heston, or quadratic rough Heston; predicting returns for **Bitcoin or ETH** using an **LSTM** or another neural network, with daily data obtained from Yahoo Finance; studying the **Root** and **Rost** optimal-stopping problems; studying the **Stefan problem** for melting ice; solving the singular free-boundary problem arising from the Black–Scholes model with **transaction costs**; or learning optimal trading strategies under **price impact** or **transaction costs** using recurrent neural networks.

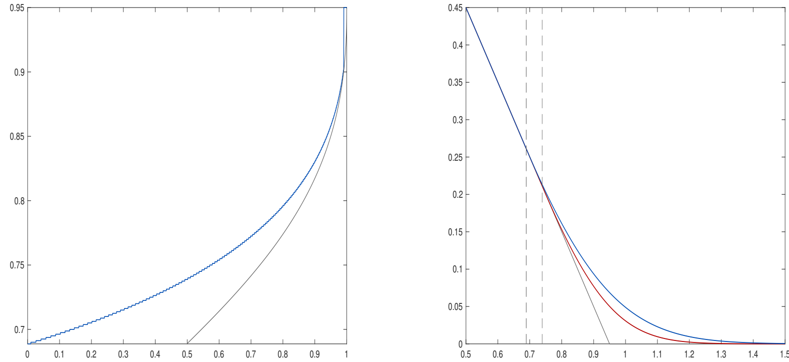

Figure 3: On the left we have plotted the optimal exercise boundary $B(t)$ in blue, obtained from an explicit finite-difference scheme for the test parameters above with one million time steps, 1600 spatial steps, $S_{\min} = 0$, and $S_{\max} = 2$, against the asymptotic expression for $B(t)$ in grey. The asymptotic expansion works well only for small values of τ. On the right we plot the American put value $P(S,t)$ for $t = 0$ in blue and $t = .5$ in red as a function of S. The critical stock prices are $B(0) = .6887$ and $B(.5) = .7388$, corresponding to the left and right vertical dashed lines. The price function for $t = .5$ and $T = 1$ is the same as the time-zero price function for maturity $T = .5$, so the red curve also represents $P(S,0)$ for $T = \frac{1}{2}$. Both curves are obtained from a single run of the explicit finite-difference scheme.

1.6 Quick proof of the scaling relation

Suppose that

$$dS_t \quad = \quad rS_t\,dt + \sigma S_t^\beta\,dW_t, \qquad S_0 > 0.$$

For $Y_t = \lambda S_t$,

$$dY_t \quad = \quad rY_t\,dt + \sigma\lambda^{1-\beta}Y_t^\beta\,dW_t, \qquad Y_0 = \lambda S_0.$$

Thus spatial scaling multiplies both the spot and strike by λ, multiplies the absolute option value by λ, and replaces σ by $\lambda^{1-\beta}\sigma$:

$$P(\lambda S_0, \lambda K, T, r, \lambda^{1-\beta}\sigma, \beta) \quad = \quad \lambda P(S_0, K, T, r, \sigma, \beta). \tag{11}$$

Now let Z solve

$$dZ_t \quad = \quad \frac{r}{\alpha}Z_t\,dt + \lambda^{1-\beta}\alpha^{-\frac{1}{2}}\sigma Z_t^\beta\,dW_t, \qquad Z_0 = \lambda S_0,$$

and define $\widetilde{W}_t = \alpha^{-1/2}W_{\alpha t}$. Then $\widetilde{W}$ is a Brownian motion, and the time-changed process $Z_{\alpha t}$ satisfies

$$dZ_{\alpha t} \quad = \quad rZ_{\alpha t}\,dt + \lambda^{1-\beta}\sigma Z_{\alpha t}^\beta\,d\widetilde{W}_t.$$

Hence $Z_{\alpha t}$ has the same law as Y_t. Under the correspondence $\tau = \alpha\theta$, stopping times $0 \le \tau \le \alpha T$ for Z correspond to stopping times $0 \le \theta \le T$ for Y, and

$$e^{-(r/\alpha)\tau} \quad = \quad e^{-r\theta}.$$

Therefore

$$P(\lambda S_0, \lambda K, \alpha T, \frac{r}{\alpha}, \lambda^{1-\beta}\alpha^{-\frac{1}{2}}\sigma, \beta) \quad = \quad P(\lambda S_0, \lambda K, T, r, \lambda^{1-\beta}\sigma, \beta). \tag{12}$$

Combining (11) and (12) gives (8).

1.7 Milstein scheme for Monte Carlo and weak convergence rates

For a general one-dimensional SDE of the form

$$dX_t \quad = \quad \mu(X_t)dt \,+\, \sigma(X_t)dW_t$$

the higher order Milstein scheme for simulating X is as follows:

$$X_{t+\Delta t} \quad = \quad X_t \,+\, \mu(X_t)\Delta W_t \,+\, \sigma(X_t)\Delta W_t \,+\, \frac{1}{2}\sigma'(X_t)\sigma(X_t)((\Delta W_t)^2 - \Delta t)$$

where $\Delta W_t = W_{t+\Delta} - W_t$. The first three terms on the right hand side are the usual **Euler scheme**, and the final term is the Milstein term. Euler and Milstein schemes both have weak order of convergence rate of 1, i.e.

$$\mathbb{E}(f(S_T)) - \mathbb{E}(f(S_T^N)) \quad = \quad O(\frac{1}{N})$$

where S_T^N is the Monte Carlo approximation to S_T with N time steps, but one would expect the error to be smaller using the Milstein scheme. (note this also includes sample variance, we are assuming in this analysis that the MC can be run an infinite number of times to obtain $\mathbb{E}(f(S_T^N)))$. See `http://people.maths.ox.ac.uk/gilesm/talks/giles_module6.pdf` by Mike Giles for a nice short discussion on error rates for Monte Carlo schemes